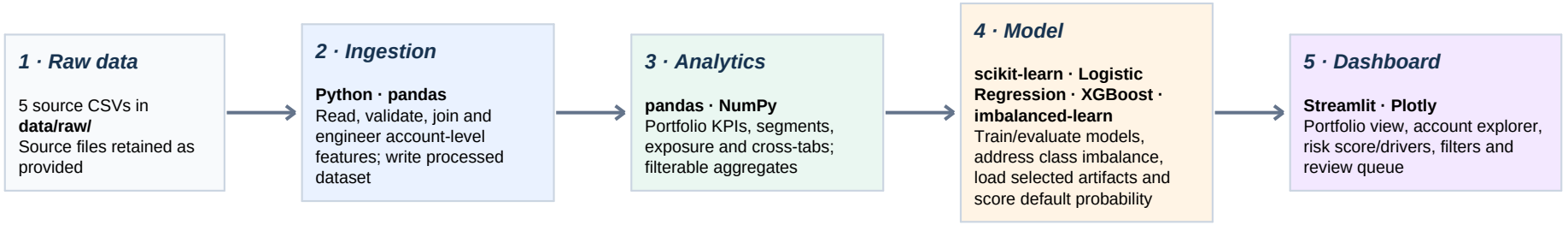
Credit Risk Portfolio Analytics — Architecture

End-to-end data flow, model scoring, dashboard, and both AI features



AI feature A · Ask the Portfolio

Trigger: analyst submits a natural-language question.
LLM call: sends the question plus a compact set of relevant computed portfolio aggregates/labels—not raw account rows or identifiers. LLM returns a concise grounded answer; unsupported questions are declined.

AI feature B · Anomaly review

Trigger: detection runs in the pipeline; flagged accounts appear in the review queue.
Detection: Python/project detector logic compares repayment/bureau patterns with history or peers; reason and priority are attached. **No LLM call.** Analyst confirms/dismisses; decision is saved.

LLM boundary & privacy

Only Ask the Portfolio calls the LLM. Sent: user question + minimized, pre-aggregated facts needed to answer. Not sent: raw CSVs, full account records, direct identifiers, model artifacts, or analyst decisions. API key comes from app secrets/environment; without it, this feature is unavailable.

Libraries: Python standard library; pandas (dataframes/transformations); NumPy (numerical operations); scikit-learn (Logistic Regression, model evaluation/scoring); XGBoost (gradient-boosted model); imbalanced-learn (class-imbalance handling); Streamlit (UI/caching); Plotly Express and Graph Objects (charts).